CNN IMAGE CLASSIFICATION

Output Screenshots – CIFAR-10 Dataset

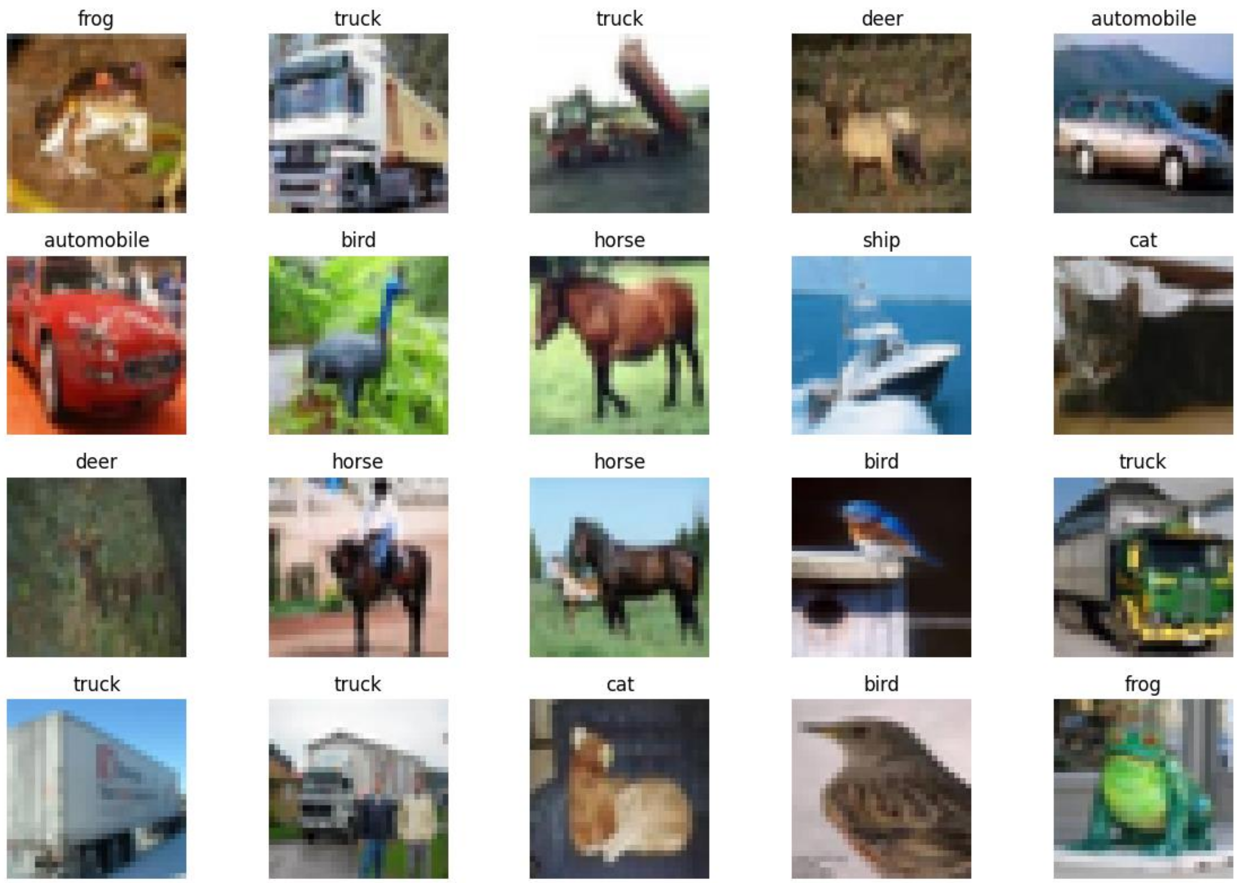
Task 1: Building an Image Classification Model using CNN

Screenshot 1: Dataset Loading Output

Colab Output

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 1830s 11us/step
Training images shape: (50000, 32, 32, 3)
Training labels shape: (50000, 1)
Testing images shape: (10000, 32, 32, 3)
Testing labels shape: (10000, 1)
```

Screenshot 2: Sample CIFAR-10 Images



Screenshot 3: Preprocessing and Data Splitting

Colab Output

```
Minimum pixel value: 0.0
Maximum pixel value: 1.0

Training data: (45000, 32, 32, 3)
Validation data: (5000, 32, 32, 3)
Testing data: (10000, 32, 32, 3)
```

Screenshot 4: CNN Model Summary

Colab Output

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 32, 32, 32)	896
max_pooling2d (MaxPooling2D)	(None, 16, 16, 32)	0
conv2d_1 (Conv2D)	(None, 16, 16, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 8, 8, 64)	0
conv2d_2 (Conv2D)	(None, 8, 8, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 4, 4, 128)	0
flatten (Flatten)	(None, 2048)	0
dense (Dense)	(None, 128)	262,272
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

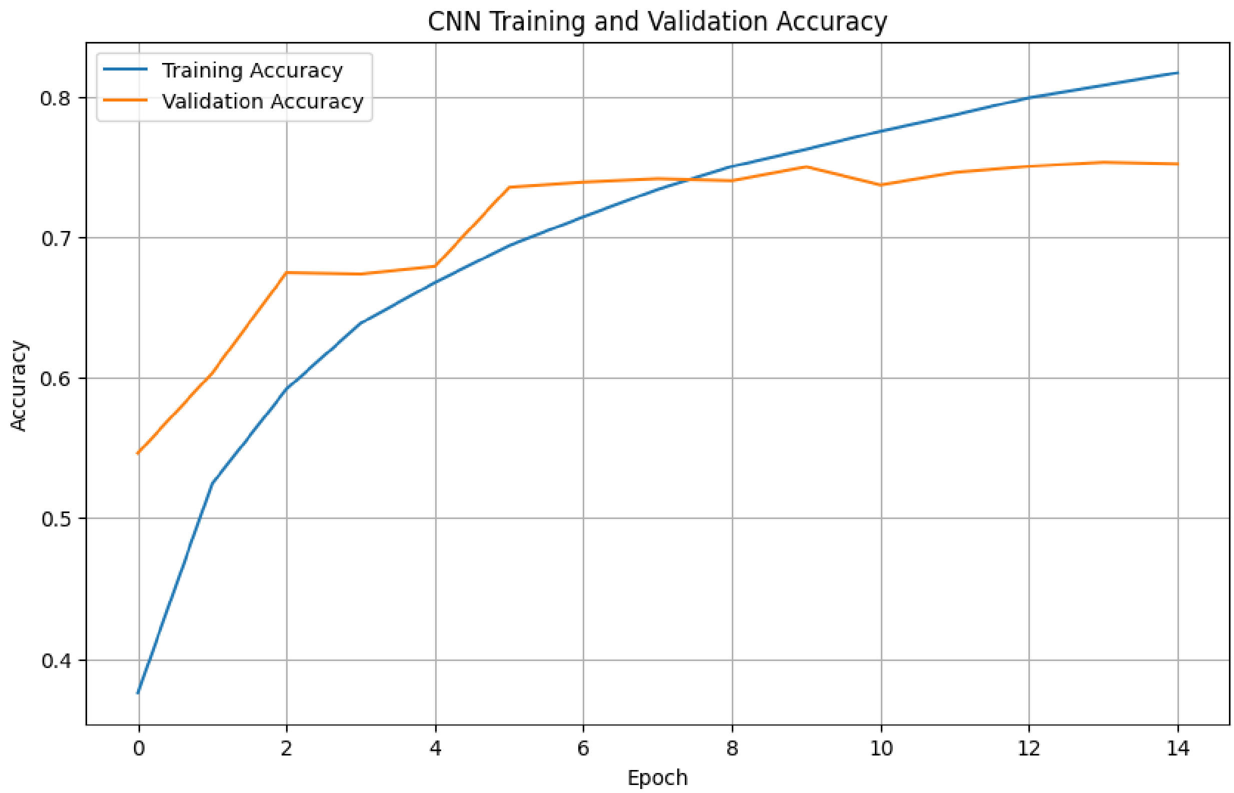
```
Total params: 356,810 (1.36 MB)

Trainable params: 356,810 (1.36 MB)

Non-trainable params: 0 (0.00 B)
```

Screenshot 5: Training Output

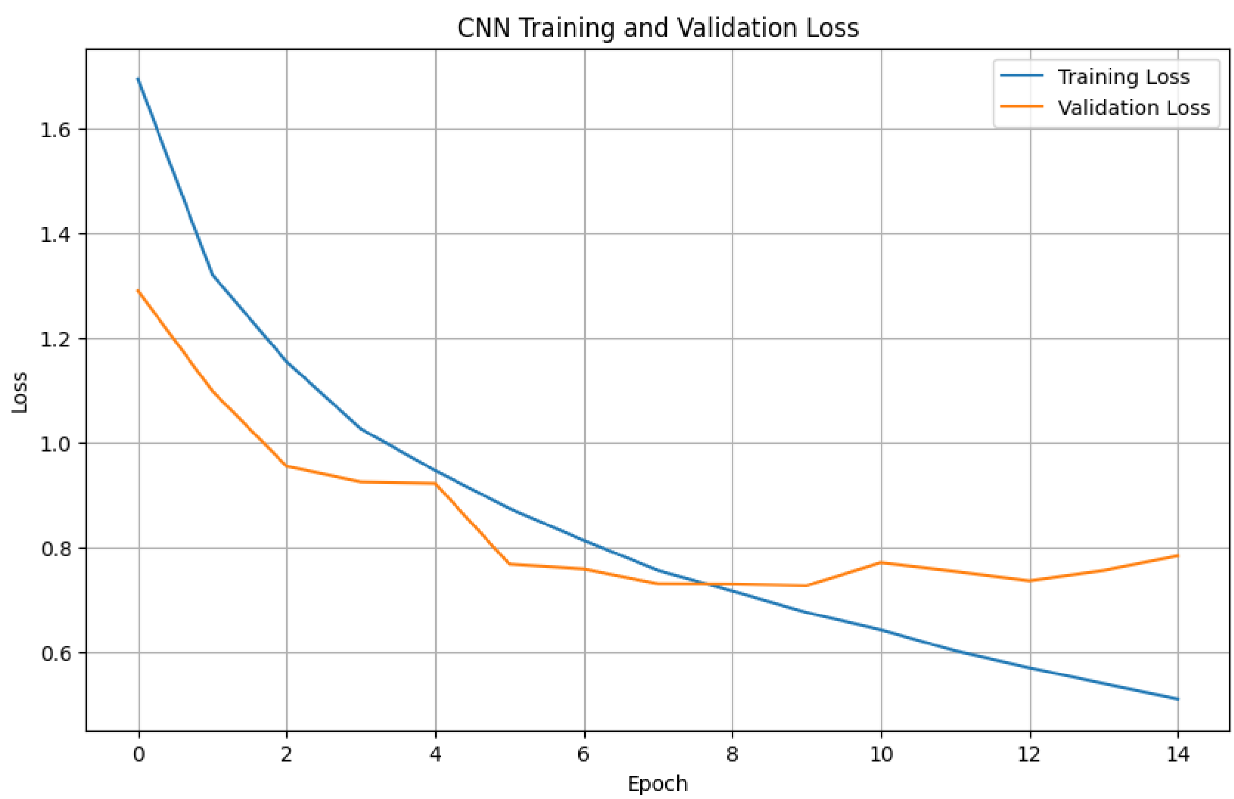
Colab Output

```
Epoch 1/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 13s 11ms/step - accuracy: 0.3757 - loss: 1.6935 - val_accuracy: 0.5462
- val_loss: 1.2899
Epoch 2/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.5245 - loss: 1.3209 - val_accuracy: 0.6032 -
val_loss: 1.0989
Epoch 3/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.5917 - loss: 1.1548 - val_accuracy: 0.6746 -
val_loss: 0.9555
Epoch 4/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.6386 - loss: 1.0267 - val_accuracy: 0.6736 -
val_loss: 0.9256
Epoch 5/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.6676 - loss: 0.9473 - val_accuracy: 0.6790 -
val_loss: 0.9229
Epoch 6/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.6939 - loss: 0.8748 - val_accuracy: 0.7354 -
val_loss: 0.7686
Epoch 7/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.7144 - loss: 0.8141 - val_accuracy: 0.7390 -
val_loss: 0.7595
Epoch 8/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 6ms/step - accuracy: 0.7339 - loss: 0.7568 - val_accuracy: 0.7414 -
val_loss: 0.7314
Epoch 9/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.7501 - loss: 0.7173 - val_accuracy: 0.7400 -
val_loss: 0.7307
Epoch 10/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.7623 - loss: 0.6766 - val_accuracy: 0.7498 -
val_loss: 0.7278
Epoch 11/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.7752 - loss: 0.6435 - val_accuracy: 0.7370 -
val_loss: 0.7715
Epoch 12/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.7867 - loss: 0.6034 - val_accuracy: 0.7458 -
val_loss: 0.7549
Epoch 13/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.7991 - loss: 0.5711 - val_accuracy: 0.7502 -
val_loss: 0.7372
Epoch 14/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 3s 5ms/step - accuracy: 0.8078 - loss: 0.5409 - val_accuracy: 0.7532 -
val_loss: 0.7567
Epoch 15/15
704/704 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - accuracy: 0.8167 - loss: 0.5117 - val_accuracy: 0.7520 -
val_loss: 0.7845
```

Screenshot 6: Accuracy Curve



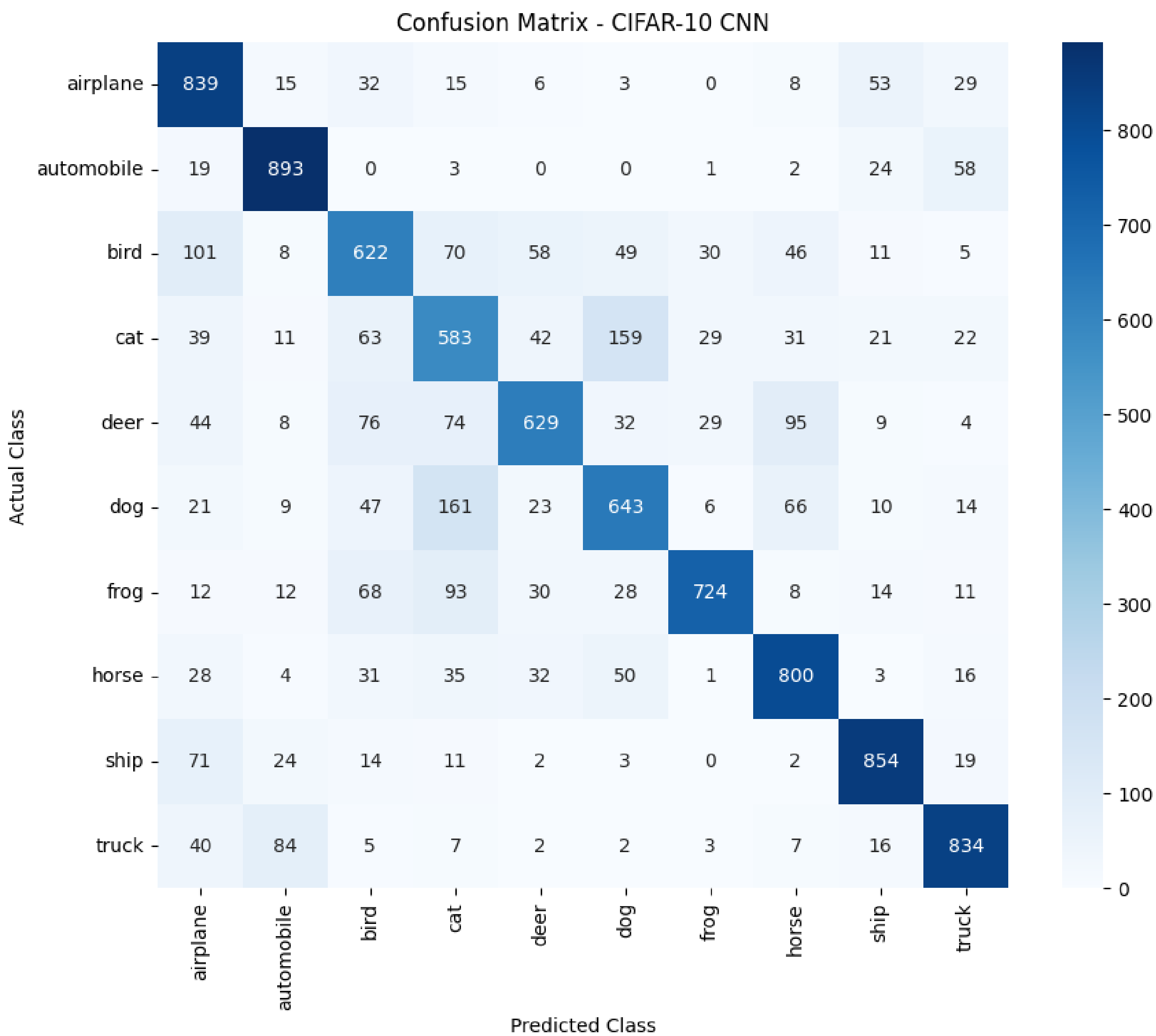
Screenshot 7: Loss Curve



<u>**Screenshot 8: Test Accuracy and Evaluation**</u>

```
Colab Output

313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - accuracy: 0.7421 - loss: 0.8440
--------------------------------
Test Loss: 0.8439542055130005
Test Accuracy: 0.7421000003814697
Test Accuracy (%): 74.21000003814697
--------------------------------
```

<u>**Screenshot 9: Confusion Matrix**</u>

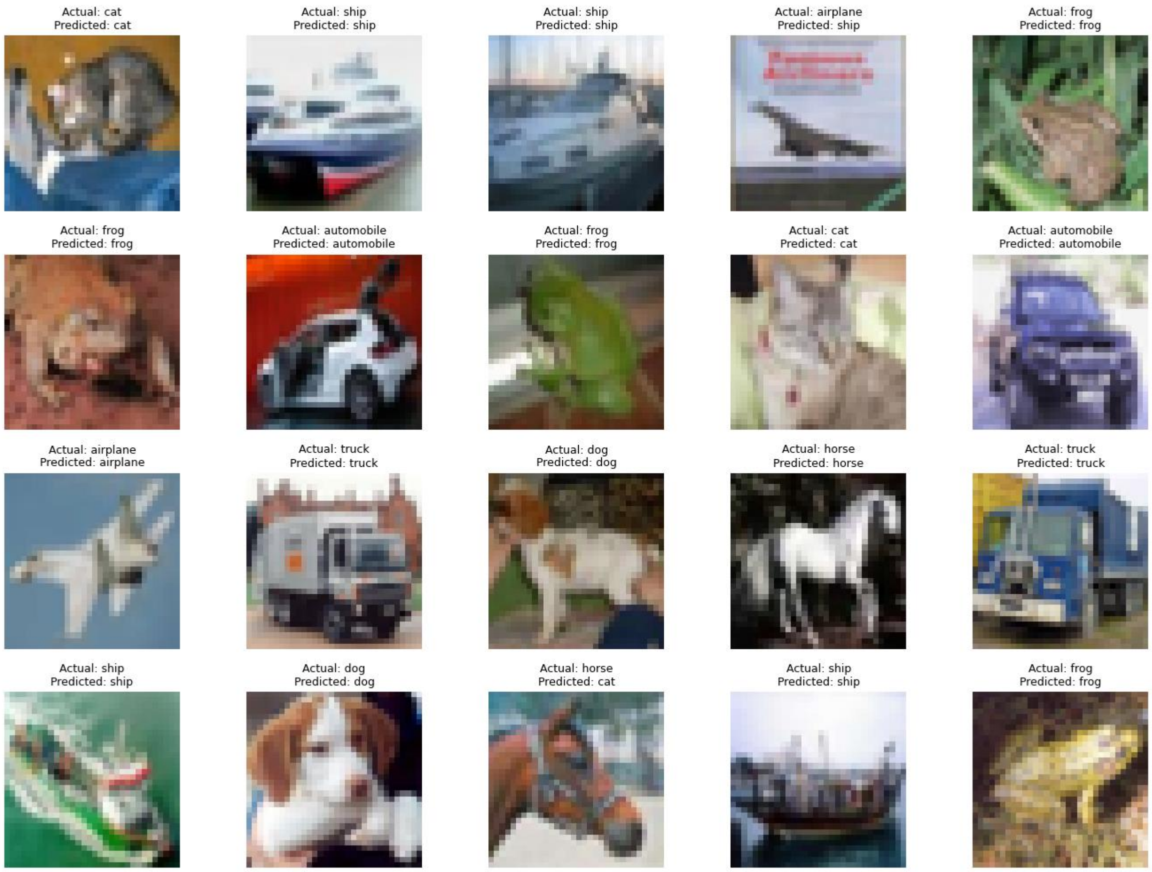


Screenshot 10: Classification Report

Colab Output

```
  precision    recall  f1-score   support

    airplane       0.69      0.84      0.76      1000
  automobile       0.84      0.89      0.86      1000
        bird       0.65      0.62      0.64      1000
         cat       0.55      0.58      0.57      1000
        deer       0.76      0.63      0.69      1000
         dog       0.66      0.64      0.65      1000
        frog       0.88      0.72      0.79      1000
       horse       0.75      0.80      0.77      1000
        ship       0.84      0.85      0.85      1000
       truck       0.82      0.83      0.83      1000

    accuracy                           0.74     10000
   macro avg       0.75      0.74      0.74     10000
weighted avg       0.75      0.74      0.74     10000
```

Screenshot 11: Actual vs Predicted Images



Screenshot 12: Saved Model Confirmation

Colab Output

```
CNN model saved successfully!
File name: cifar10_cnn_model.keras
```